

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Discrete Mathematics, Calculus III

PROJECTS

Autonomous B2B Deal-Sourcing Agent | *Python, LangGraph, LangSmith, Docker, Celery, Pydantic* github.com/Verdent06/Gen-Agentic-Pipeline

Dockerized LangGraph pipeline for private equity acquisition sourcing — deployed across 4 live theses for search funds and mid-market PE clients

- Engineered a LangSmith eval pipeline with 20 adversarial business archetypes — manufacturers, SaaS tools, PE holding companies, and geographic mismatches the LLM otherwise passed — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.
- Engineered Node 3 as a pure-Python triangulated consensus gate — no LLM calls — that computes a 0–100 lead score by fuzzy-matching the pipeline query, state business registry, and live website crawl in a three-way weakest-link check, then hard-fails leads on legal status, industry, geography, or independence before the score threshold applies.
- Built Node 5 as a fully deterministic succession-risk scoring engine — no LLM calls — that infers PE acquisition readiness from heuristic signals (founder operating status, PE backing, website stagnation, years in operation) into a configurable weighted 0–100 composite, then enriches each passing company with Hunter.io contact data before CSV export.
- Packaged a Dockerized LangGraph execution pipeline as a recurring B2B lead-gen product, cold-approached and closed a paid pilot with a consulting firm, and deployed Redis/Celery workers to autonomously aggregate and deliver qualified leads on a scheduled basis.

EXPERIENCE

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Eliminated a 27% audio upload failure rate by building fault-tolerant RxJS logic that detects expired S3 presigned URLs and regenerates them mid-flight, and handles MIME-type negotiation for raw WAV files that Angular's HTTP client rejected silently.
- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated the silent-audio problem in a voice-AI product — most frames sent to the Whisper backend were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.
- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.

TECHNICAL SKILLS

Languages Python, TypeScript

Frameworks & Cloud .Flask, Angular, RxJS, LangGraph, LangSmith, Pydantic, AWS, Docker, Celery, Redis, Git

AI / ML Silero VAD, ONNX Runtime, Pandas

Tools Web Workers